

## **C. E. "Dallas" Hoffman**

### **Geospatial Analyst, GISP, MGIST**

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- GISP - Official GISCI GIS Professional Certification (#162180)
- MGIST - Masters in Geospatial Information Science and Technology
- 10 years of experience in the GIS space
- Bounded appointments in both private, public (grant-based) sectors
- Experience, knowledge over a broad range of roles - from technician to consultant

#### *SKILLS SUMMARY*

GIS Platforms: (ESRI Suite) ArcGIS Online, ArcGIS Pro, ArcServer, GRASS GIS, QGIS, Experience Builder

Programming & Scripting: Python, R, HTML, CSS, SQL, GitHub, Jupyter Lab/Notebook, Visual Studio

Data Management: MS Access, PostgreSQL, pgAdmin, MS Server, ArcGIS Server Manager, ArcSDE

#### *EXPERIENCE*

##### **GIS Analyst II and Data Science Consultant** '25 - '26

@ Pickett and Associates, an ESP Company – Raleigh, NC

Power Business Unit - Transmission Engineering

- Consulted with client engineers to establish cartographic and web mapping (Arc Portal) needs utilizing survey, aerial, environmental, transportation, AutoCAD, and Google Earth datasets.
- Identified project needs, developed data models, advanced spatial and data analysis workflows using LiDAR-derived datasets, and presented solutions and recommended next steps to project stakeholders.
- Authored ArcGIS Experience Builder web apps, including custom widgets, for client-facing interactivity.

##### **Geospatial Data Science Consultant, Project Principal** '23 - '24

@ North Carolina State University – Raleigh, NC

Department of Civil, Construction, and Environmental Engineering

Coastal Hazards Engineering Research Lab

- Served as project principal and geospatial data consultant for the NC Department of Transportation's Coastal Monitoring Program (CMP) wherein I generated geostatistical intel datasets, graphs, charts, and map layouts to support critical coastal hazard, erosion mitigation, and shoreline engineering operations.
- Engaged in strategic, focused, monthly meetings with the CMP project consulting engineer to QA/QC data analysis outputs, photogrammetry interpretations, and feature creation to ensure data accuracy and precision.
- Innovated automation tools such as consolidated geoprocessing tasks in Arc Model Builder and Arc(Py) Notebook that led to streamlined data analysis workflows, improved efficiency, and reduced the potential for introduction of human error.
- Updated all outdated ArcMap procedures to ArcPro compatibility throughout the CMP project ops manual because I had intel that NC State ArcMap licenses would terminate in June; I achieved this goal in rapid time which resulted in eliminating the department's need to purchase costly individual licenses for CMP.

- Managed and maintained the Coastal Engineering Research Lab on Centennial Campus of roughly ten graduate students and visiting scholars.
- Planned and led bi-weekly meetings for the lab's principal professor, Dr. Anarde.

#### **Principal Data Science Engineer and Geospatial Consultant** '22 - '23

@ North Carolina State University – Raleigh, NC

Department of Biological and Agricultural Engineering

Precision Agriculture Data Science Extension

Approached by the N.C. State Climate Office to develop an open source, cloud-based data analysis tutorial component for a U.S. Soy/United Soybean Board funded research project.

- Engaged in strategic consultation with Extension Specialists for needs-gathering operations leading to project goals determination.
- Authored and deployed both a statistical and geospatial data analytics interactive, cloud-based, live-coding environment notebook.
- Programmed using several coding languages – R, Python, Bash (magics).
- Forked, adapted, and applied a bioinformatics, polyglot Jupyter coding notebook environments framework package to program interchangeably/interoperably.
- Staged yield monitor data geovisualizations in a public-facing ArcGIS web map application with data analytics dashboard for user-friendly access and presentation.
- Incorporated external data layers through REST services to provide further contextual information to gain valuable insights.

#### **Research Analyst and GIS Subject Matter Expert** '19 - '23

@ Zonda (formerly Hanley Wood Meyers Research) – Remote

Data Intelligence

- Served as the sole GIS, Geospatial subject matter expert on an 11+ member research team within the Data Intelligence department covering all 50 states.
- Conducted research using state, county, and local GIS open data resources, database systems, satellite imagery, property records, deeds, plats, maps, consolidated real estate, and assessor tax roll data.
- Updated the proprietary national database of dozens of specific real estate market growth performance metrics and data intelligence points for some 20+ major industry development and builder clients in 50+ counties among 10 states.
- Developed to fill a previously undiscovered void in internal research project management, a semi-programmed methodologies and resources system for efficacious temporal project performance, progress, and productivity tracking for project metrics of 50+ counties in ten states.
- Accomplished – with the implementation of the aforementioned – a significant increase of ~50% in output productivity with a simultaneous decrease of similar amount in input operations.

#### **Geospatial Consultant (end), GIS Technician (begin)** '16 - '19

@ Theorem Geo Associates – Charlotte, NC

- **Lead Project Developer and Manager**

**Client: Duke Energy Land Strategy and Real Estate (RE)**

- Consulted with client – Duke Energy GIS Director – to define scope and effort for contract project to address business needs.
- Independently designed and developed from the ground up, deployed, and maintained a sustainable, collaborative online data management system for major, continuous RE project involving complex acquisitions, QA/QC tasks, organization, extracting/transforming/loading

- (ETL), and processing of quarterly-updated geospatial parcel data for Duke Energy's seven-state, 260+ county service territory.
- Established sustainable and automated workflows, standards, ETL practices, and processes. As a result, a \$250,000 four-year contract with renewal option was awarded by Duke Energy to the project, as its value was realized to internal enterprise mapping operations utilized company-wide from RE and capital projects to utility corridor vegetation management and easements mapping.
- Deliverables of GDBs, SDEs, and other spatial data layers as requested by clients utilizing project outputs.
- **Geospatial Data Analyst**  
**Client: Duke Energy Environmental Health and Safety (EHS)**
  - Synthesized, performed QA/QC tasks, and organized large amounts of tabular data and GDBs from the NC Department of Natural Resources – Natural Heritage Program.
  - Prepared map layer deliverables to Transmission Vegetation Management Department for Best Management Practices (BMP).
- **Geospatial Data Analyst**  
**Client: Georgia Power (Southern Company) Emergency Storm Services (ESS)**
  - Assisted with the preparation and QA/QC processes of different aspects of spatial data and relational databases relevant to four Storm Staging Sites for Georgia Power's ESS.
- **Data Project Designer**  
**Client: Duke Energy Transmission Vegetation Management (TVM)**
  - Performed major data synthesis, organization, and joining of historical annual Transmission Line Work Orders to a web map app and relational database I established to drastically help improve TVM record cohesion and gaining valuable performance insights from the information created.

## *EDUCATION*

### **Master of Geospatial Information Science and Technology (MGIST) '23**

North Carolina State University, Raleigh, NC  
 Special Designation Professional Science Masters (PSM)

### **Graduate Certificate in Geographic Information Systems (GIS) '21**

North Carolina State University, Raleigh, NC  
 Issued by the Center for Geospatial Analytics, College of Natural Resources

### **Bachelor of Arts, International Studies, Concentration in EU Environmental Affairs '12**

University of North Carolina Charlotte | Minors in Political Science, History, and German

## *CERTIFICATIONS*

- GIS Certification Institute GIS Professional • NASA TOPS Open Science Mission
- Inclusive Excellence Program • Elsevier Research Peer Reviewer
- U.S. Department of Homeland Security – FEMA Emergency Management Institute (EMI)
  - Applications of GIS for Emergency Management
  - Geospatial Information Systems Specialist
  - Geospatial Information Infrastructure
- U.S. Department of Homeland Security – Cybersecurity & Infrastructure Security Agency (CISA)
  - Critical Infrastructure Protection
  - Incident Management Response: Ransomware Attacks